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BUPT Compilers Lab6 2024Fall

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1 Code

1.1 json.h

Json's data structure is implemented in json.h.

1.1.1 JsonValue is the basic data structure of Json. It contains a JsonType and a JsonUnion.

```
struct JsonValue {  
    JsonType type;  
    JsonUnion value;  
};
```

1.1.2 JsonType is an enumeration of all possible types of Json.

```
typedef enum {  
    JSON_NULL,  
    JSON_BOOLEAN,  
    JSON_INTEGER,  
    JSON_NUMBER,  
    JSON_STRING,  
    JSON_ARRAY,  
    JSON_OBJECT  
} JsonType;
```

1.1.3 JsonUnion is a union of all possible values of Json.

```
typedef union {  
    int boolean;  
    long long integer;  
    double number;  
    char *string;  
    JsonArray array;  
    JsonObject object;  
} JsonUnion;
```

1.1.4 sonArray is a dynamic array of JsonValues.

```
typedef struct {  
    size_t size;  
    size_t capacity;  
    JsonValue *values;  
} JsonArray;
```

1.1.5 JsonObjectMember is a key-value pair of JsonValues.

```
typedef struct {  
    char *key;  
    JsonValue *value;  
} JsonObjectMember;
```

1.1.6 JsonObject is a dynamic array of JsonObjectMembers.

```
typedef struct {  
    size_t size;  
    size_t capacity;  
    JsonObjectMember *members;  
} JsonObject;
```

1.2 json.c

json.c as follows.

```
#include "json.h"  
#include <stdio.h>  
  
void json_print(JsonValue json) {  
    switch (json.type) {  
        case JSON_NULL:  
            printf("null");  
            break;  
        case JSON_BOOLEAN:  
            printf(json.value.boolean ? "true" : "false");  
            break;  
        case JSON_INTEGER:  
            printf("%lld", json.value.integer);  
            break;  
        case JSON_NUMBER:  
            printf("%g", json.value.number);  
            break;  
        case JSON_STRING:  
            printf("%s", json.value.string);  
            break;  
        case JSON_ARRAY:  
            printf("[");  
            for (size_t i = 0; i < json.value.array.size; i++) {  
                if (i > 0) {  
                    printf(", ");  
                }  
                json_print(json.value.array.values[i]);  
            }  
            printf("]");  
            break;  
        case JSON_OBJECT:  
            printf("{");  
            for (size_t i = 0; i < json.value.object.size; i++) {  
                if (i > 0) {
```

```

        printf(", ");
    }
    printf("\"%s\": ", json.value.object.members[i].key);
    json_print(*json.value.object.members[i].value);
}
printf("}");
break;
default:
    break;
}
}
}

```

1.3 main.c

The availability of Json's data structure is tested in main.c as follows.

```

#include <stdio.h>
#include <stdlib.h>
#include "json.h"

int main() {
    // Create a JSON object
    JsonValue json;
    json.type = JSON_OBJECT;
    json.value.object.size = 0;
    json.value.object.capacity = 10;
    // Allocate memory for members
    json.value.object.members = malloc(sizeof(JsonObjectMember) *
json.value.object.capacity);
    // Add a member of type string
    json.value.object.members[json.value.object.size].key = "Name";
    json.value.object.members[json.value.object.size].value = malloc(sizeof(JsonValue));
    json.value.object.members[json.value.object.size].value->type = JSON_STRING;
    json.value.object.members[json.value.object.size].value->value.string = "项枫";
    json.value.object.size++;
    // Add a member of type integer
    json.value.object.members[json.value.object.size].key = "Student ID";
    json.value.object.members[json.value.object.size].value = malloc(sizeof(JsonValue));
    json.value.object.members[json.value.object.size].value->type = JSON_INTEGER;
    json.value.object.members[json.value.object.size].value->value.integer = 2022211570;
    json.value.object.size++;
    // Add another member of type integer
    json.value.object.members[json.value.object.size].key = "Class";
    json.value.object.members[json.value.object.size].value = malloc(sizeof(JsonValue));
    json.value.object.members[json.value.object.size].value->type = JSON_INTEGER;
    json.value.object.members[json.value.object.size].value->value.integer = 2022211801;
}

```

```
    json.value.object.size++;  
    // Print the JSON object  
    json_print(json);  
    puts("");  
    return 0;  
}
```

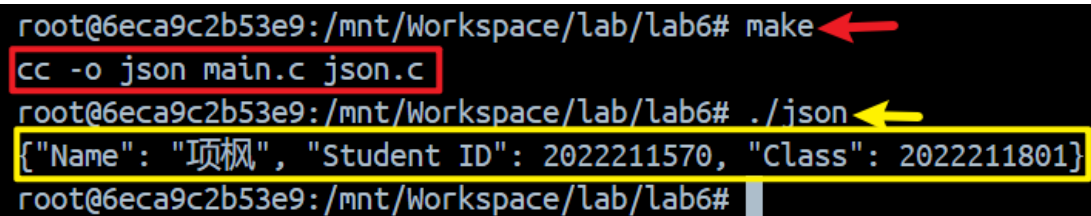
2 Run

These commands were used:

```
make lltest
```

```
./json
```

The screenshot of the compilation and execution results.



```
root@6eca9c2b53e9:/mnt/Workspace/lab/lab6# make  
cc -o json main.c json.c  
root@6eca9c2b53e9:/mnt/Workspace/lab/lab6# ./json  
{"Name": "项枫", "Student ID": 2022211570, "Class": 2022211801}
```

The screenshot shows a terminal window with the following commands and output:
1. `make` (indicated by a red arrow) compiles the program, with the command `cc -o json main.c json.c` highlighted in a red box.
2. `./json` (indicated by a yellow arrow) runs the program, with the output `{"Name": "项枫", "Student ID": 2022211570, "Class": 2022211801}` highlighted in a yellow box.